

Q:- Find fcn's domain, range, boundary of fcn's domain, determine if the domain is open or closed or neither & decide if domain is bounded or unbounded. Also sketch the domain ^{& shade}. Also find interior & boundary points.

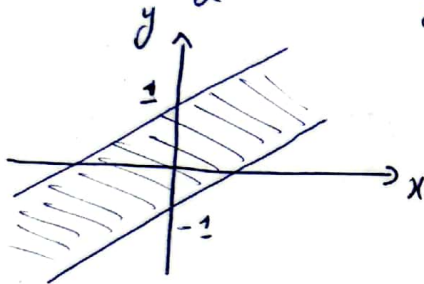
(a) $f(x, y) = \sin^{-1}(y-x)$

Sol As $-1 \leq y-x \leq 1$

$$-1+x \leq y \leq 1+x$$

Domain: $\{(x, y) \in \mathbb{R}^2 \mid -1 \leq y-x \leq 1\}$

Range: $-\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$



All the pts lying b/w lines $-1+x$ & $1+x$ are Interior pts.

= = = = on the lines $-1+x$ & $1+x$ are boundary pts.

Domain is closed as the boundary is included. Domain is unbounded because we can't enclose it in circle of finite radius. The lines $-1+x$ & $1+x$ are boundary of fcn's domain

(b) $f(x, y) = \arcsin(x^2+y^2-2)$

Sol As $-1 \leq x^2+y^2-2 \leq 1$

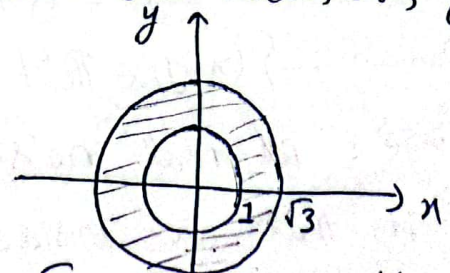
$$-1+2 \leq x^2+y^2 \leq 1+2$$

$$1 \leq x^2+y^2 \leq 3$$

Domain is the annulus centered at origin with outer radius $=\sqrt{3}$ & inner radius $=1$.

Domain: $\{(x, y) \in \mathbb{R}^2 \mid 1 \leq x^2+y^2 \leq 3\}$

Range: $-\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$



All the pts lying b/w circles of radius 1 & $\sqrt{3}$ are Interior pts.

All the pts lying on the circle of radius 1 & $\sqrt{3}$ are boundary pts. Domain is closed as the boundary is included. Domain is bounded because we can enclose it in the circle of finite radius. The circle $x^2 + y^2 = 1$ & $x^2 + y^2 = \sqrt{3}$ are boundary of fn's domain.

(c) $f(x, y) = \arcsin(2x + y)$

Sol

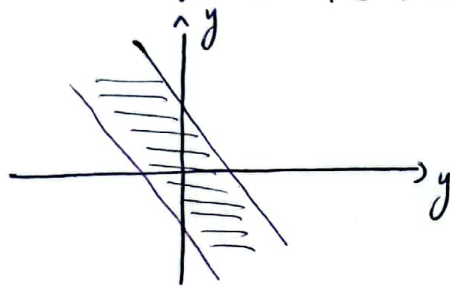
$$-1 \leq 2x + y \leq 1$$

$$-1 - 2x \leq y \leq 1 - 2x$$

Domain: $\{(x, y) \in \mathbb{R}^2 \mid -1 - 2x \leq y \leq 1 - 2x\}$

Range: $-\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$

Domain is the set of all pts in xy -plane which are b/w the lines $y = -2x - 1$ & $y = -2x + 1$, including pts on lines.



All the pts lying b/w lines $-1 - 2x$ & $1 - 2x$ are interior pts

= = = = on lines " = = = boundary pts

Domain is closed because the boundary is included. Domain is unbounded we can't enclose it with circle of finite radius. The lines $-1 - 2x$ & $1 - 2x$ are boundary of fn's domain.

(d) $f(x, y) = \frac{\sin(xy)}{x^2 + y^2 - 25}$

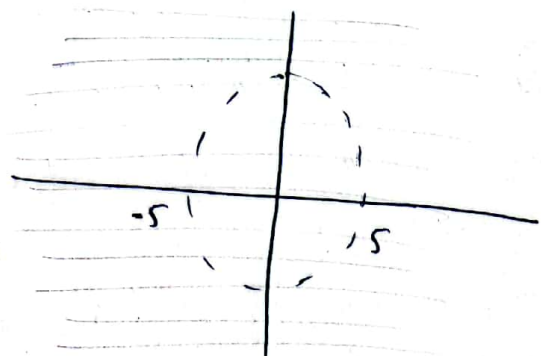
Sol

$$x^2 + y^2 - 25 \neq 0$$

$$x^2 + y^2 \neq (5)^2$$

Domain: $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \neq (5)^2\}$

Range: All real no's.



The pts that are inside & outside the circle of radius 5 are interior pts.

The pts on the circle of radius 5 acts as domain's boundary. The domain is open as it doesnot contain B.P's. The boundary pts are the pts that lie exactly on the circle of radius 5. & the domain is unbounded.

$$(e) f(x,y) = \frac{\sin^{-1}(x^2-y)}{\sqrt{y}}$$

Sol

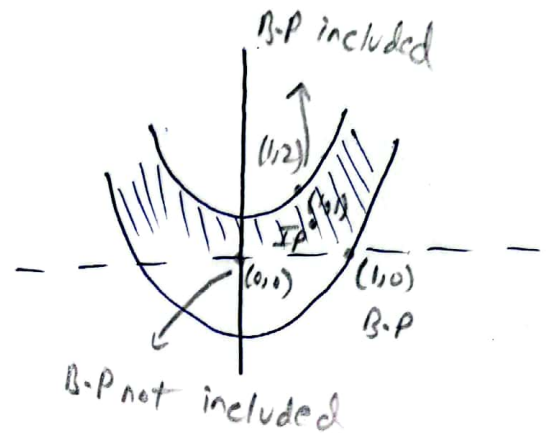
$$\begin{aligned} -1 &\leq x^2 - y \leq 1 \\ -1 - x^2 &\leq -y \leq 1 - x^2 \\ 1 + x^2 &\geq y \geq -1 + x^2 \end{aligned}$$

$$\text{For } \sqrt{y} \quad y > 0$$

$$\text{So } y \leq 1 + x^2$$

$$y \geq x^2 - 1$$

$$y > 0$$



$$\text{Domain: } \{(x,y) \in \mathbb{R}^2 \mid 1+x^2 \geq y \geq -1+x^2, y > 0\}$$

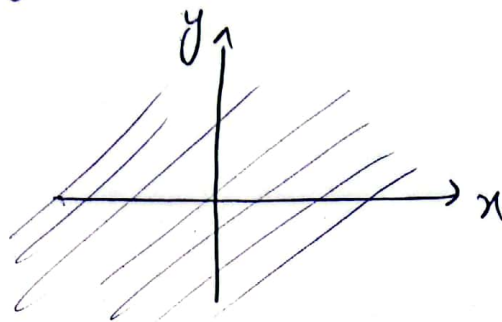
Range: $[-\pi/2, \pi/2]$, with actual values attained depending on y . For large y , ftn values will be close to 0.

This region is neither open nor close because some B.P's are included & some are not. Region is unbounded. The boundary includes the line $y=0$ (excluding origin) & curves defined by $1+x^2$ & $-1+x^2$, for $y > 0$.

$$(f) h(x,y) = \frac{1}{1+x^2+y^2}$$

$$\text{Sol } D: \text{Entire plane } (\mathbb{R}^2)$$

$$R: (0, 1]$$



Every ordered pair in $\mathbb{R}^2$ is interior pt. As there is no edge to this region so there are no B.P's. Region is both open & close. It is unbounded. It does not have a boundary.

Q3:- A paint store carries two brands of latex paint. Sales figures indicate that if the first brand is sold for x_1 dollars per gallon & second for x_2 dollars per gallon, the demand for the first brand will be $D_1 = 200 - 10x_1 + 20x_2$ gallons per month & demand for second brand will be $D_2 = 100 + 5x_1 - 10x_2$ gallons per month. Express the paint store's total revenue from the sale of the paint as a fn of prices.

Sol Total revenue from both brands of paint per month

$$R(x_1, x_2) = (\text{No. of 1st brand's paint sold})(\text{Price per gallon}) + (\text{No. of 2nd brand's paint sold})(\text{Price per gallon})$$

$$R(x_1, x_2) = D_1 x_1 + D_2 x_2$$

$$R(x_1, x_2) = (200 - 10x_1 + 20x_2)(x_1) + (100 + 5x_1 - 10x_2)(x_2)$$

$$R(x_1, x_2) = 200x_1 - 10x_1^2 + 20x_1x_2 + 100x_2 + 5x_1x_2 - 10x_2^2$$

$$R(x_1, x_2) = 200x_1 - 10x_1^2 + 100x_2 - 10x_2^2 + 25x_1x_2$$

Q6:- A right cylinder with height y & radius x is used to store wheat. The top & bottom of the cylinder has the shape of hemisphere with radius x . It is desired to represent the vol as a fn of 2 variables.

Sol

$$V = V_{\text{upper hemisphere}} + V_{\text{cylinder}} + V_{\text{lower hemisphere}}$$

$$V_{\text{upper hemisphere}} + V_{\text{lower hemisphere}} = V_{\text{sphere}}$$

$$V = V_{\text{sphere}} + V_{\text{cylinder}}$$

$$V = \frac{4}{3} \pi x^3 + \pi x^2 y$$

$$V = \pi x^2 \left(\frac{4}{3} x + y \right)$$